



Measurements of the deformation zone around a split-axis snow micropenetrometer tip

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ABSTRACT

Current physical models for the snow micropenetrometer (SMP) make the assumption that the tip of the SMP probe only induces stress or fractures in the ice grains in direct contact. However, past research indicates the existence of an extended deformation zone around the tip which contains many strained or fractured ice grains. A noncontact optical strain measurement system directly observed snow deformation within well rounded snow caused by a split-axis SMP probe simulator. Results confirm the existence of a deformation zone ahead of and normal to the surfaces of the SMP probe tip. By assuming axial symmetry, the estimated volume of the deformation zone was on the order of 1000 mm³. Approximately 40%–50% of this volume was due to deformation in front of the tip, which is currently not considered in micromechanical penetration resistance theories for snow.

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1. Introduction

Accurate data on snowpack stratigraphy is critical for avalanche forecasting. To this end, manual snow pit profiles in combination with stability tests are recorded in the field to observe snow stratigraphy and evaluate the strength of potential weak layers (e.g. Schweizer and Jamieson, 2007; van Herwijnen and Jamieson, 2007). Such manual snow cover data are time consuming and prone to subjective interpretation.

Cone penetrometers have therefore been developed to measure force resistance through the depth of a snowpack (e.g. Dowd and Brown, 1986). A widely used penetrometer is the snow micropenetrometer (SMP; Schneebeli and Johnson (1998), Schneebeli et al. (1999)) which can take full depth profiles of snow hardness in minutes and is not subject to observer bias. Force measurements are taken from a 5 mm diameter, 60° probe tip at intervals of 4 μm and are interpreted through either empirical or physically-based models (e.g. Havens et al., 2012; Johnson and Schneebeli, 1999; Löwe and van Herwijnen, 2012; Marshall and Johnson, 2009). Past research has shown that the SMP can be used to distinguish layers with different characteristics, for instance abrupt changes in hardness or grain type (e.g. Pielmier and Schneebeli, 2003).

Physical theories have neglected compaction of snow grains around the tip of the SMP probe (e.g. Johnson and Schneebeli, 1999), arguing that compaction is negligible in low density snow. Penetrometry research in porous soils has shown compaction ahead of the tip to be a factor (Fritton, 1990). For the SMP in snow, Johnson (2003) theorizes that compaction ahead of the tip is negligible and assumes compaction only

exists normal to the tip surface. However, previous work has revealed the existence of a deformation zone ahead of the tip of probes of various shapes as they are driven through snow (Floyer and Jamieson, 2006, 2010; Gleason and Marshall, 2006; Riche and Schneebeli, 2010; van Herwijnen (2013)). The deformation zone is defined as the region where grains of snow have experienced any deformation, which could be due to elastic strain or bond breakage.

In this study, we describe a series of experiments designed to directly observe and measure the deformation zone around a split-axis SMP probe model with tip using ARAMIS, a high-resolution optical strain measurement system using particle image velocimetry (PIV). Well-sintered, small-grain snow was chosen because it is dense and therefore represents the greatest potential for discrepancy from theories that assume negligible compaction.

2. Methods

2.1. Experimental procedure

Experiments were conducted to measure the deformation zone ahead of and normal to the SMP tip. Given the conical shape of the SMP tip, it was assumed that deformation of snow around the SMP probe is axisymmetric (Floyer, 2008), and deformation in the radial direction could be directly observed through penetration of a split-axis probe through snow similar to the work of Gleason and Marshall (2006). A split-axis probe with dimensions similar to the SMP probe and tip was driven in the snow with an SMP motor. This model was designed to provide geometric similarity while allowing for optical strain measurements during penetration. The transition between the tip and the probe was not as abrupt for the split-axis probe as for the actual

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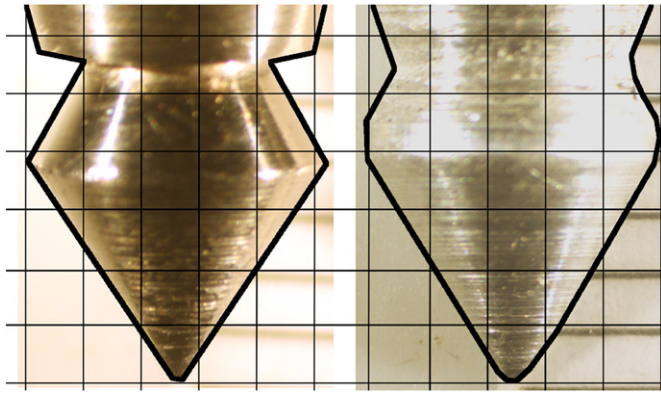


Fig. 1. Comparison of actual SMP tip (left) and split-axis replica used in the experiments (right). 1 mm grid.

SMP (Fig. 1). Since the split-axis probe is a geometric model only, snow resistance was not measured.

To prepare snow samples for the experiment, dendritic snow crystals made previously in a lab snow breeder were collected. Snow was sieved into a $26.5 \times 58 \times 46$ cm box and allowed to sinter overnight. The box's plexiglass front window was removed and the snow surface was speckled with paint by dipping a brush in paint and flicking paint onto the snow surface. Paint specks were generally interconnected. This technique provided contrast for the particle image velocimetry (PIV) analysis. The window was replaced after the paint dried. The replica probe was attached to the SMP rod and placed against the plexiglass. Penetration was then initiated by the SMP at a rate of 20 mm/s (normal operating speed). Eight tests were performed in snow with properties at test time specified in Table 1.

2.2. Particle image velocimetry

Strain was measured using ARAMIS 3D 5M, an optical deformation analysis system from GOM industries (<http://www.gom.com/>) which measures three dimensional displacement fields under static or dynamic loading. Noncontact optical measurements are used to determine parameters such as material coordinates, displacements, speed, acceleration, strain, strain rate and material characteristics during loading. Strain ranging from 0.05% to over 100% can be measured.

Strain is calculated using a PIV algorithm to track the displacement of patterns in the images. The software calculates strain in overlapping square windows of pixels. Images were 2448×2050 pixels with spatial resolution of 0.08 mm/pixel as computed automatically from ARAMIS calibration. In the analysis, 12×12 pixel windows with 33% overlap were used. Images were recorded at 15 frames per second, yielding measurements at 1.33 mm penetration increments. By tracking the movement of each paint speck, the software calculates the 2D strain tensor in each window. Minor principal strain magnitude and direction were calculated automatically from the eigenvalues and eigenvectors,

respectively. Minor principal strain was chosen to analyze the deformation zone because it represents the highest magnitude of compressive strain in the snowpack (compressive strain is negative).

ARAMIS is unable to track rigid body motion of individual grains; rather it calculates relative motion between specks of paint to find strain. Excessive rigid body motion of individual grains can create invalid deformation patterns where a continuum strain calculation is no longer valid. Strain then becomes incalculable, but deformation is still occurring. Blurring in the images due to high rates of deformation can also create a zone where ARAMIS cannot calculate strain. When ARAMIS fails to report strain, it is unknown whether this is due to excessive rigid body motion or due to image blur. However, rigid body motion and blurring are both indicative of deformation and therefore zones of incalculable strain are always in a deformation zone.

Prior to SMP penetration, a random noise threshold was established as the highest magnitude of minor principal strain measured prior to penetration. Noise thresholds were always less than or equal to 1%, but did vary from test to test (Table 1). Only minor principal strains with greater magnitude than the noise threshold were considered to be in the deformation zone.

2.3. Deformation zone size parameters and image processing

The deformation zone was characterized by three parameters: its two dimensional area, width at the level of the widest part of the SMP tip, and length ahead of the tip (Fig. 2). To automate calculation of these parameters, the location of the probe tip must be known in all images. The probe tip was speckled with a paint pattern and the ARAMIS software was able to track a point on the probe that is a known distance above the tip.

The deformation zone A was defined as the area (mm^2) normal to or ahead of the tip that displayed strain greater than the noise threshold. To calculate the area of the deformation zone, a color overlay showing the minor principal strain was binarized to show the deformation zone in white and undeformed snow in black (Fig. 3). Then, based on the known location of the SMP probe tip, the deformation zone area was calculated by counting the deformed pixels and multiplying by the pixel area. The area was subdivided into two regions as illustrated in Fig. 2. The area highlighted in green represents deformation normal to the tip and the area highlighted in red represents deformation ahead of the tip.

The height, H , of the deformation zone was measured as the vertical distance (mm) directly beneath the SMP tip that displayed minor principal strain with magnitude greater than the noise threshold. The width, W , was measured as the sum of the horizontal distances (mm) left and

Table 1
Test properties. Grain types according to Fierz et al. (2009).

Test #	Noise threshold	Density (kg/m^3)	Grain size (mm)	Grain type
1	1%	203 kg/m^3	<1 mm	RG
2	1%			
3	0.6%			
4	0.6%			
5	1%	257 kg/m^3	<1 mm	RG
6	0.9%			
7	0.9%			
8	0.9%			

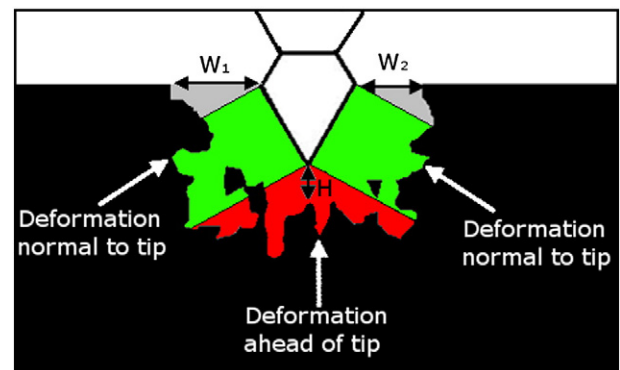


Fig. 2. Sample illustration of the dimensions of the deformation zone derived from the binarized image. Tip normal deformation is green and deformation ahead of the tip is red.

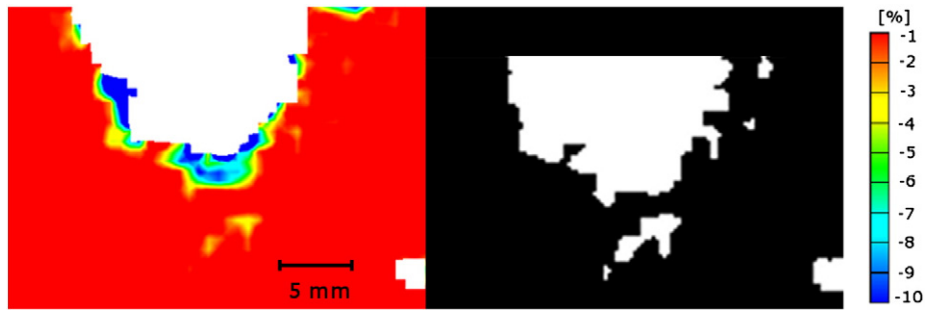


Fig. 3. Example strain field before and after binarization. The binarized image is cut off above the widest part of the SMP tip as it was presumed that snow above that level would not influence force reading. Only the contiguous deformation zone around the tip is counted.

right of the widest part of the SMP tip that displayed minor principal strain of magnitude greater than the noise threshold. As shown in Fig. 2,

$$W = W_1 + W_2. \quad (1)$$

To calculate the width of the deformation zone automatically, the row of pixels at the widest part of the penetrometer cone tip is located in each image. Pixels are then counted at each side of the SMP tip until a strain value below the noise threshold is encountered. Total counted pixels are multiplied by scale (mm/pixel) to calculate W .

The volume of the deformation zone is calculated based on the area of deformation. An axisymmetric deformation zone is assumed because the rounded, sintered grains used in these experiments tend to have small anisotropy. Although the deformation zone does not appear axially symmetric in many images, this assumption can still be useful to obtain first order volume estimates. Thus, the volume of a full cylindrical SMP deformation zone can be estimated by a volume integral in polar coordinates based on revolving the deformation zone area:

$$V = \int_A \int_0^\pi r dA d\theta \quad (2)$$

where dA is a differential area, r is the distance from probe's symmetry axis to the differential area, and θ is the rotation angle about the axis. By taking each square pixel in the image to be a differential area, the volume represented by a pixel is:

$$V_{\text{pixel}} = A_{\text{pixel}} t \pi. \quad (3)$$

The volumes represented by each pixel, when added, comprise the estimated deformation zone volume. As with area, volume is subdivided into a region normal to the tip and a region ahead of the tip.

3. Results

Results showed a deformation zone around the SMP tip in every test. There was always a zone of incalculable strain, due to either rigid body motion or blurring in images, surrounded by a zone of calculable strain. The deformation zone extended ahead of and to the side of the tip, as well as to the side of the larger rod driving the tip (Fig. 4).

3.1. Snow deformation

Fig. 4 shows an example of the deformation zone from test 2. The area with the contoured overlay represents areas with calculable strain; strain was incalculable in other regions. Each image typically exhibited a zone immediately surrounding the tip where strain could not be calculated. Around this zone, there was a zone showing measurable strain which diminished with distance from the SMP tip. It should be noted that due to blurring associated with low frame capture rate, the software had difficulty in processing strain in the tip. Therefore there is no strain overlay on the tip in many images. It is also likely that deformation in

snow near the tip is also incalculable due to blurring. Further from the tip, near where strain could be calculated, the images appear less blurry and likely represent an invalid strain condition due to rigid body motion.

3.2. Deformation zone height

On average, H ranged from 3.7 to 5.4 mm, consistent with the results of van Herwijnen (2013) and Gleason and Marshall (2006). During testing, H fluctuated rapidly as the probe moved through the snow (see results of five selected tests in Fig. 5).

3.3. Deformation zone width

The width of the deformation zone was generally greater than the height. In tests 2, 4, and 6, W increased for the first 15 to 30 mm of penetration, then decreased and reached a steady state value after about 40 mm (Fig. 6). In tests 1, 3, 5, 7, and 8, W increased and then reached the steady state value without peaking first (Fig. 7). On average, W ranged from 15.1 to 18.0 mm, larger than values reported by van Herwijnen (2013).

3.4. Deformation zone area

Similar to the deformation zone width, the deformation zone area generally increased, but in one of two patterns. For tests 1, 2, 4, and 6, the area of the deformation zone increased for the first 15 to 30 mm of penetration, then decreased and reached a steady state (Fig. 8). For tests 3, 5, 7 and 8, the area of the deformation zone approached steady state without a noticeable peak (Fig. 9). In all tests, median total area was between 83 and 110 mm² as shown in Fig. 10.

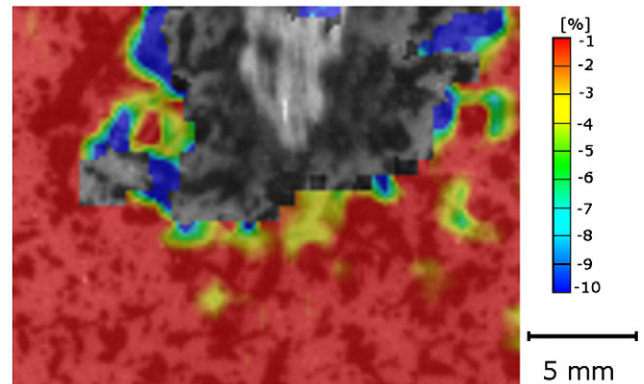


Fig. 4. Contour plot of minor principal strain overlaid on sample image of SMP tip through snow. Images consistently showed a zone of incalculable strain surrounded by gradually decreasing measurable strain. Negative strain denotes compression.

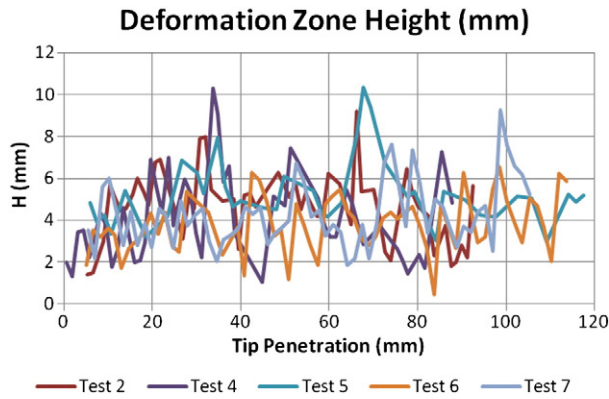


Fig. 5. Height of deformation zone with penetration depth for five selected tests.

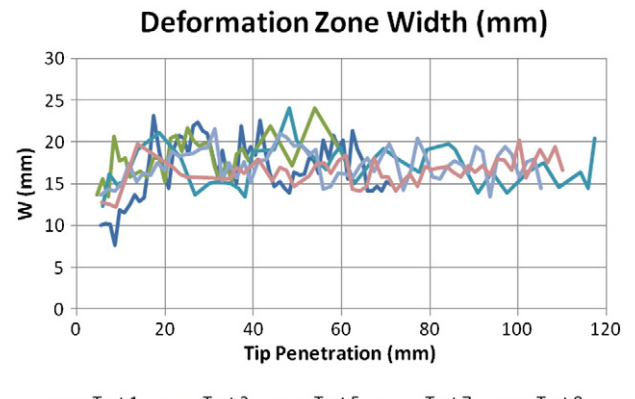


Fig. 7. Deformation zone width for tests 1, 3, 5, 7 and 8.

3.5. Deformation zone volume

As with area, a steady state volume was reached (Fig. 11) after approximately 40 mm. Median volumes were on the order of 800 to 1300 mm³ as seen in the quartile box plots in Fig. 12. Although the deformation zone had a large mean volume, there was a considerable fluctuation between sequential images. It was common for volume to vary by 10% or more between images (Fig. 13). To estimate the significance of force feedback caused by deformation ahead of the tip, the volume of the deformation zone ahead of the tip was compared to the volume of the deformation zone normal to the tip. In all our tests, the median portion of the deformation zone ahead of the tip ranged from 48% to 57% of total deformed volume. Fig. 14 shows deformation ahead of the tip as a percent of total deformation. Unlike the total volume, the percentage did not climb to equilibrium.

3.6. Minor principal strain direction

Principal strain orientations were also calculated during SMP penetration. Minor principal strain was chosen because it shows the direction in which major compaction occurs. Minor principal strain was aligned approximately normal to the SMP probe tip, indicating that the tip is the primary cause of deformation (Fig. 15) as assumed in previous theory (Johnson and Schneebeli, 1999; Marshall and Johnson, 2009). Force effects of compaction are a function of minor principal strain orientation. Only the vertical component (sine of cone half angle) of minor principal strain will contribute to force reading in the SMP tip.

4. Discussion

This series of eight experiments confirms the existence of a deformation zone that extends normal to and ahead of a replica SMP tip, as seen in similar experiments with other probes (Floyer and Jamieson, 2010; Riche and Schneebeli, 2010) and with a full SMP probe (van Herwijnen, 2013). In general, W was about 3 times the width of the SMP tip and H was about equal to the height of the tip. Total deformation zone area was about 10 times the area of the SMP tip and estimated volume assuming axial symmetry was on the order of 1000 mm³.

During penetration, the SMP probe is inserted in the snow and a deformation zone builds around the tip. As individual bonds break in the deformation zone, rigid body motion ensues until a critical density is reached and force reaches a maximum (Johnson, 2003). Our results showing a gradual increase in the size of the deformation zone until roughly 40 mm of penetration seem to corroborate this idea, but some tests showed no gradual buildup. The observed size of H suggests deformation ahead of the tip that could affect the accuracy of narrow layer delineation (van Herwijnen, 2013). Although there were no clear overall trends in H as a function of penetration depth, constant fluctuation of H suggests a continual buildup and breakdown of a plug of snow in front of the SMP tip that will affect force feedback, particularly at layer transitions. The observed fluctuation is consistent with Johnson's (2003) model, which predicts development of a maximum average penetration force subject to fluctuations caused by random roughness and mechanical events.

The SMP was specifically designed with a cone angle of 30° to “minimize the compaction of snow and the formation of a snow body ahead of the penetrometer” (Johnson and Schneebeli, 1998). However, deformation zone volumes in these tests reveal deformation

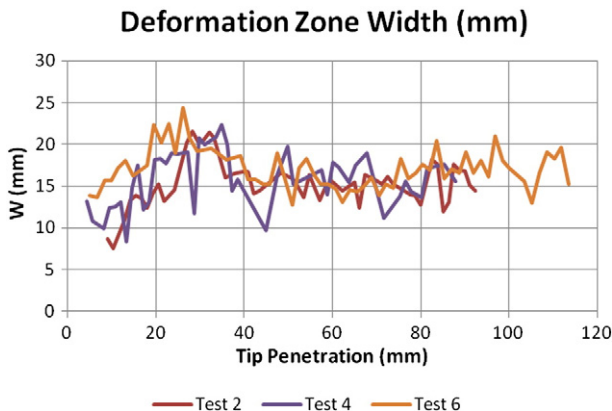


Fig. 6. Deformation zone width with penetration depth for tests 2, 4 and 6.

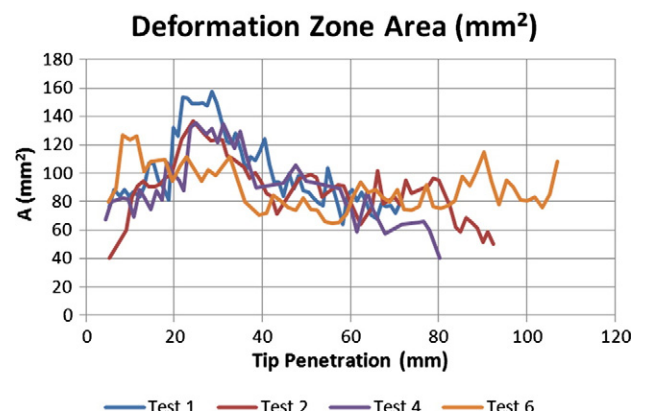


Fig. 8. In tests 1, 2, 4 and 6 area peaked and then decreased to steady state.

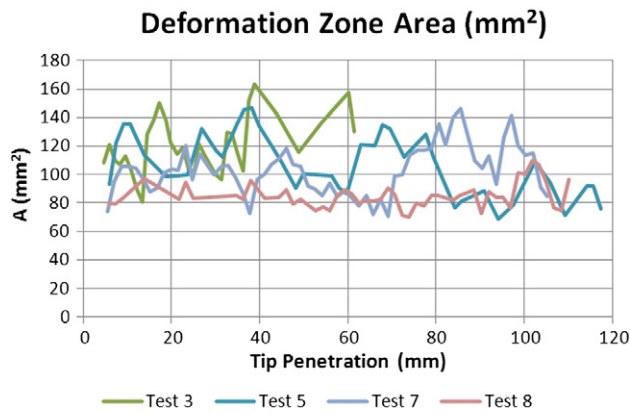


Fig. 9. In tests 3, 5, 7 and 8, steady state was reached without peaking first.

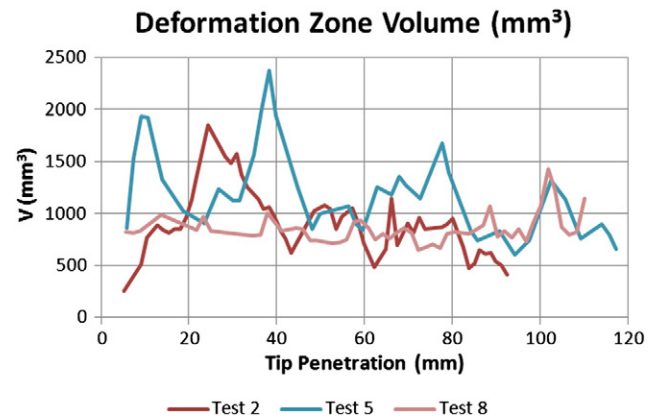


Fig. 11. Volume of the deformation zone with penetration depth for tests 2, 5 and 8.

ahead of the penetrometer to be on the order of 40%–50% of the total deformation volume. Deformation ahead of the tip aligns directly with the force sensor, while deformation force normal to the tip surface will only contribute an axial component of the force contributing to deformation. With a 30° cone angle, only half the force ($\sin(30^\circ) = 0.5$) that deforms snow normal to the tip will contribute to force feedback, meaning that volumetric deformation ahead of the tip will disproportionately affect recorded force feedback. Given that deformation ahead of the tip is nearly equal in volume to normal deformation, and that normal deformation should influence force half as much per unit volume, deformation ahead of the tip could create more force feedback than deformation normal to the tip.

During penetration, some compaction is caused by the SMP tip attached to the force sensor, and some compaction is caused by the larger rod behind the tip. Only compaction caused by the tip should influence force feedback. For these experiments, it was assumed that only snow ahead of the widest part of the tip would influence force feedback on an actual SMP probe, and all snow behind this point on the tip would not. A demarcation between tip-influencing deformation and rod-affected deformation will be necessary before any new physical force feedback theory can account for the effects of compaction. It was assumed that minor principal strain normal to the penetrating surface indicated a compressive effect from the surface, but principal strain directions were not always clearly perpendicular to a given portion of the probe surface as shown in Fig. 15. This result suggests that not all

compaction is perpendicular to the probe surface, and forces may be redirected by microstructural elements.

Johnson (2003) uses the idea of a penetrometer effective surface (PES), the surface area of broken and compacted micro-structural elements in contact with intact snow. The PES is the surface area available to cause deformation. In our tests, the consistent zone of incalculable strain suggests the existence of a compaction zone containing highly deformed, rearranged broken particles, as described by Floyer and Jamieson (2006). Estimations of force feedback based on PES assume a conic surface in contact with undeformed snow and no deformation ahead of the tip (i.e. $H = 0$). Although the shape of the deformation zone may not be as simple as predicted, there is support in the literature for an initial force buildup followed by quasi-steady state force (Johnson, 2003) and corroborating evidence from some field tests (e.g. Johnson and Schneebeli, 1999) and laboratory experiments (van Herwijnen and Miller, 2013). The buildup and subsequent quasi-steady deformation zone sizes seen here support the idea of a force buildup in a general sense. As the deformation zone increases in size, more grains are being deformed, and therefore force feedback should increase.

The high variability in the size of the deformation zone between sequential images suggests abrupt changes in snow microstructure as the probe penetrates the snow. The high magnitude and apparently erratic nature of these changes suggest that the temporal resolution of these images was insufficient to observe behavior within the deformation

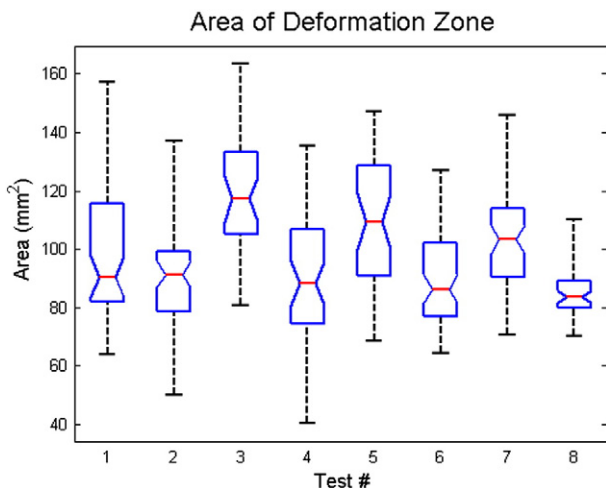


Fig. 10. Notched box plots of deformation zone visible area. Tests #1–4 are from the first set of tests, and tests #5–8 are from the second set.

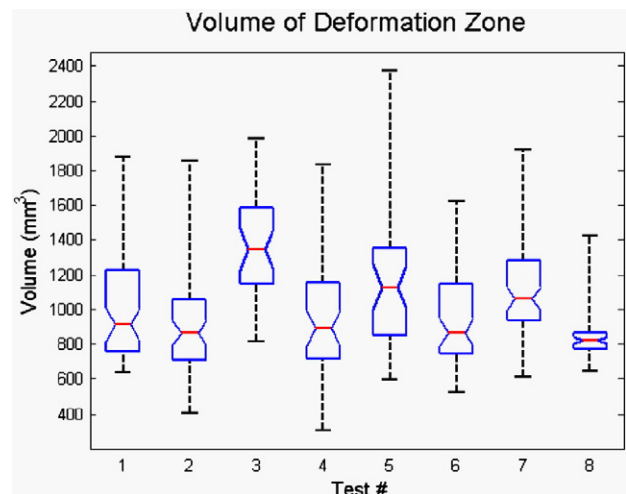


Fig. 12. Notched box plots of deformation zone estimated volume.

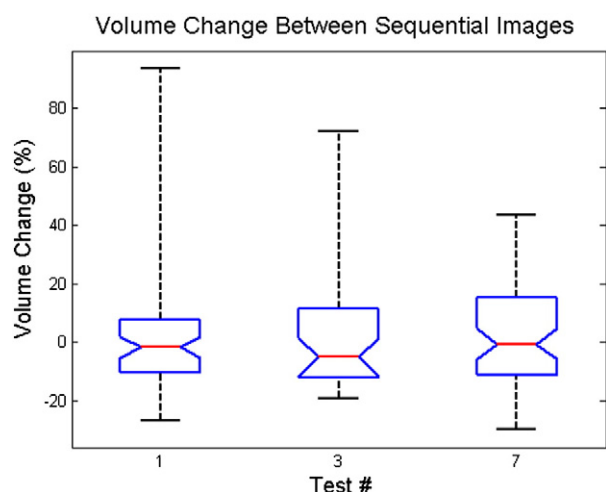


Fig. 13. Distribution of volume change between sequential image frames for tests 1, 3 and 7.

zone. Nonetheless, images did provide “snapshots” of the deformation zone development, size and progression.

5. Conclusions

Eight tests performed in well-sintered rounded snow showed that a substantial deformation zone exists around the SMP tip during penetration. This type of snow could be the slab in an avalanche, and correct interpretation of force feedback is critical. These tests represent a “worst case scenario” for SMP signal interpretation as the small, densely packed grains tested are particularly prone to compaction. These tests may not be relevant to some persistent weak layers such as surface hoar, where high porosity and vertical alignment of grains can actually yield a “flat-line” force feedback unlikely to be associated with compaction (Lutz and Birkeland, 2011). Different methods of signal interpretation may be necessary for different grain types.

The deformation zone around the SMP tip almost certainly affects force feedback as measured by the SMP. Nearly 50% of the deformation zone volume we observed was ahead of the tip and therefore not accounted for in current SMP deformation models. In order to correctly interpret SMP penetration data, accounting for this deformation zone is crucial.

Due to difficulties in tracking individual grains through PIV, a better approach to studying the force feedback of this deformation zone could be through numerical modeling of penetration through individual grains. Furthermore, high speed videography would be useful to

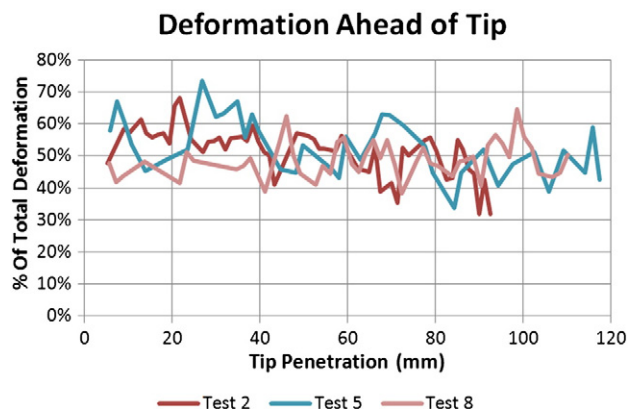


Fig. 14. Deformation ahead of the tip as a percent of total deformation volume.

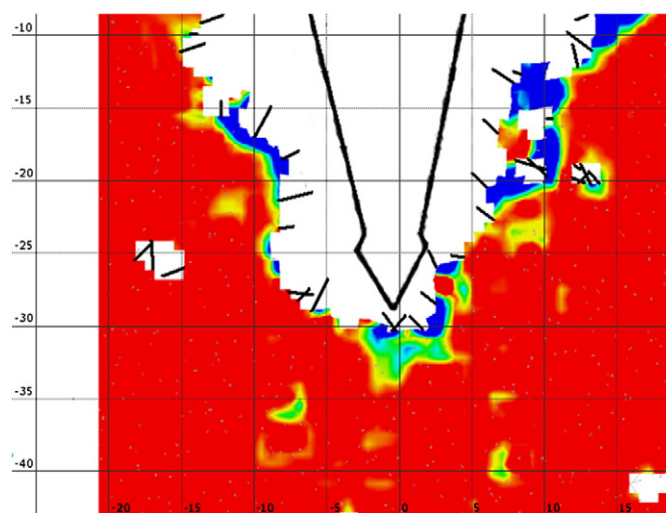


Fig. 15. Minor principal strain orientation is shown by black lines. SMP tip outline is superimposed on image. Grid in mm.

observe the rapid fluctuations in the deformation zone. These abrupt changes could be caused by the breaking of individual bonds, or by rapid fluctuation of force chains which transmit load between grains (Liu and Nagel, 1992). However, which of these phenomena occurred in our experiments is unknown.

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